Jai Rana

Robotics & Control Engineer

RESUME_1260

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Total Experience: 14 years

PROFESSIONAL SUMMARY

Robotics & Control Engineer with 14 years of experience in robotics, using ROS 2, Python, C++, MATLAB. Experienced in autonomous navigation, control loops and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: PLC, PID Control, LiDAR, MATLAB, SLAM, OpenCV, Motion Planning, Python, C++, Simulink, ROS 2, Robot Kinematics

Methods: Autonomous navigation, Control loops, Sensor fusion

PROFESSIONAL EXPERIENCE

Lead Robotics & Control Engineer | PrismArc Solutions | Jan 2021 - Present

- Lowered navigation recovery events by 33% using improved obstacle handling
- Reduced robotic path completion time by 14% through route and controller tuning
- Validated 734 robot simulation scenarios with recorded sensor logs
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Robotics & Control Engineer | Urbanaut Engineering | Jan 2017 - Dec 2020

- Validated 634 robot simulation scenarios with recorded sensor logs
- Improved pick-position repeatability by 19% after calibration updates

Associate Robotics & Control Engineer | Meridian Vector Labs | Jul 2012 - Dec 2016

- Lowered navigation recovery events by 16% using improved obstacle handling
- Reduced robotic path completion time by 16% through route and controller tuning

PROJECTS

Warehouse Mapping Trial | PID Control, LiDAR, Motion Planning, PLC

- Benchmarked LiDAR SLAM configurations under moving obstacles.
- Documented acceptance criteria and evaluated the design using control loops and sensor fusion.

Vision-Guided Pick Cell | PLC, C++, Simulink, MATLAB

- Integrated camera calibration, pick pose estimation and manipulator trajectories.
- Documented acceptance criteria and evaluated the design using control loops and sensor fusion.

EDUCATION

B.Tech in Mechatronics Engineering, Blue River College of Technology, 2012

Relevant Methods: Autonomous navigation, Control loops, Sensor fusion

CERTIFICATIONS

Industrial Robot Safety (2020)